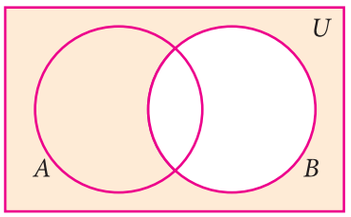
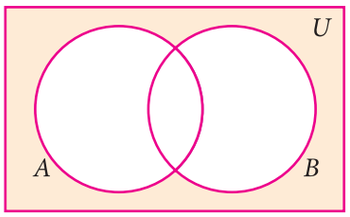


A'



B'



$A' \cap B'$

.... (2)

Fig.1.31

From (1) and (2), it is verified that $(A \cup B)' = A' \cap B'$

Example 1.26 If $U = \{x : x \in \mathbb{Z}, -2 \leq x \leq 10\}$,

$A = \{x : x = 2p+1,\ p \in \mathbb{Z}, -1 \leq p \leq 4\}$, $B = \{x : x = 3q+1,\ q \in \mathbb{Z}, -1 \leq q < 4\}$,

verify De Morgan's laws for complementation.

Solution

Thinking Corner

$A \cap (A \cup B)' =$____

Given $U = \{-2,-1,0,1,2,3,4,5,6,7,8,9,10\}$,

$A = \{-1,1,3,5,7,9\}$ and $B = \{-2,1,4,7,10\}$

Law (i) $(A \cup B)' = A' \cap B'$

Now, $A \cup B = \{-2,-1,1,3,4,5,7,9,10\}$

$(A \cup B)' = \{0,2,6,8\}$ ..... (1)

Then, $A' = \{-2,0,2,4,6,8,10\}$ and $B' = \{-1,0,2,3,5,6,8,9\}$

$A' \cap B' = \{0,2,6,8\}$ ..... (2)

From (1) and (2), it is verified that $(A \cup B)' = A' \cap B'$

Thinking Corner

$(A \cup B)' \cup (A' \cap B) =$____

Law (ii) $(A \cap B)' = A' \cup B'$

Now, $A \cap B = \{1,7\}$

$(A \cap B)' = \{-2,-1,0,2,3,4,5,6,8,9,10\}$ ..... (3)

Then, $A' \cup B' = \{-2,-1,0,2,3,4,5,6,8,9,10\}$ ..... (4)

From (3) and (4), it is verified that $(A \cap B)' = A' \cup B'$

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